Combinatorial Algorithms (Math 482), 2026 Spring   Problem Set 7

Due by Wednesday, March 11, 2026.

1. **Minimizing Delay.** We are given n jobs with durations $A[1..n]$ and deadlines $D[1..n]$ and we want to choose an ordering (permutations), so that when we run all jobs on a single machine starting from time 0, so maximum delay is minimized. In class, we discussed a greedy algorithm for this problem: Sort the jobs by nondecreasing deadline. We have also seen that there may be multiple optimal solutions.

   (a) What is the maximum number of optimal solutions over all input of size n?

   (b) What is the minimum number of optimal solutions for n jobs (for large n)?

   (c) What happens if we sort the jobs by duration? Does this ordering always give an optimal solution? Either prove that it does or find a counterexample.

2. **Minimizing Wait Time.** We are given n intervals $(R[1], Q[1]), \ldots, (R[n], Q[n])$, that represent requests for renting a classroom at CSUN. We want to accommodate all requests, but we have only one classroom available. We need to choose an ordering of the requests, and then schedule all requests in this order (back-to-back), so that we allocate $Q[i] - R[i]$ time for request i, however the actual start time $S[i]$ of request i is not necessarily $R[i]$. We want to minimize the maximum *wait time*, where the wait time of request i is defined as $S[i] - R[i]$ (that is, actual start time minus requested start time).

   Design an algorithm for this problem. Prove that it finds an optimal solution. What is the running time of your algorithm?

3. **Elementary Graph Problems.** We are given an undirected graph $G = (V, E)$ with n vertices and m edges. Design an algorithm for each of the problems below, and analyze their worst-case running times in terms of n and m.

   (a) Is the graph G a tree?

   (b) Does the graph G contain any even cycle (C_{2k} for a positive integer k)?

   (c) How many 3-cycles does G contain?

4. **Classifying Roundworms.** Your friend at a biology lab at CSUN is trying to classify n specimens of roundworms into subspecies. The biologists have already compared m pairs of roundworms: For each of these pairs (i, j), they either determined that i and j belong to the *same* subspecies or to *different* subspecies. They wonder whether their data is consistent with the hypothesis that all n specimens belong to only two subspecies, A and B. Design an algorithm that can answer this question in $O(n + m)$ time.